

Data Handling and Visualization

Lab Experiments

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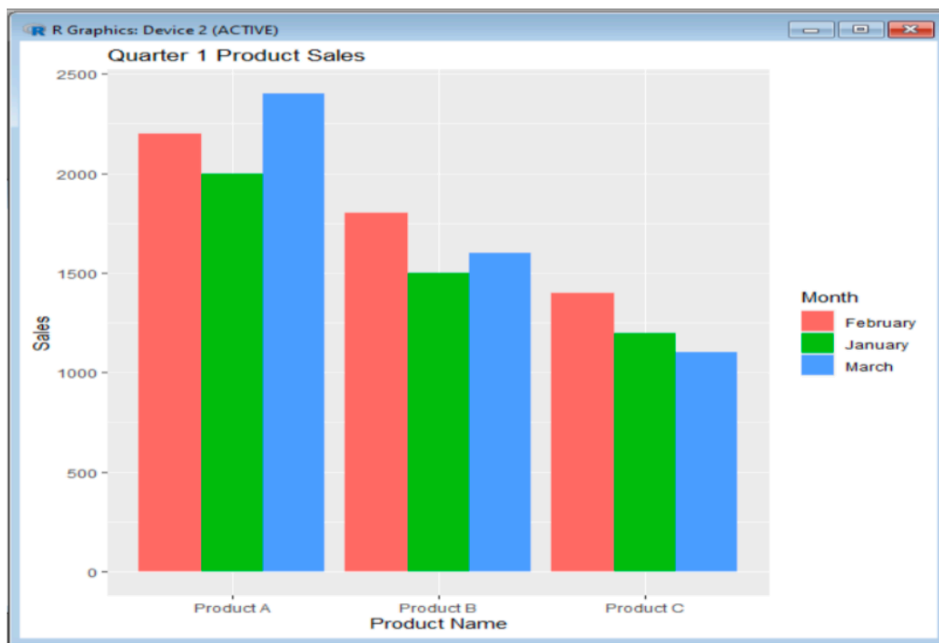
Set 6

1. Grouped bar chart to visualize the sales of each product for the first quarter:

Python code:

```
sales <- data.frame(  
  ProductID = c(1,2,3),  
  ProductName = c("Product A","Product B","Product C"),  
  January = c(2000,1500,1200),  
  February = c(2200,1800,1400),  
  March = c(2400,1600,1100)  
)  
sales_long <- pivot_longer(  
  sales,  
  cols = c(January, February, March),  
  names_to = "Month",  
  values_to = "Sales"  
)  
ggplot(sales_long,  
  aes(x = ProductName,  
  y = Sales,  
  fill = Month)) +  
  geom_bar(stat = "identity",  
  position = "dodge") +  
  labs(title = "Quarter 1 Product Sales",  
  x = "Product Name",  
  y = "Sales")
```

Output:

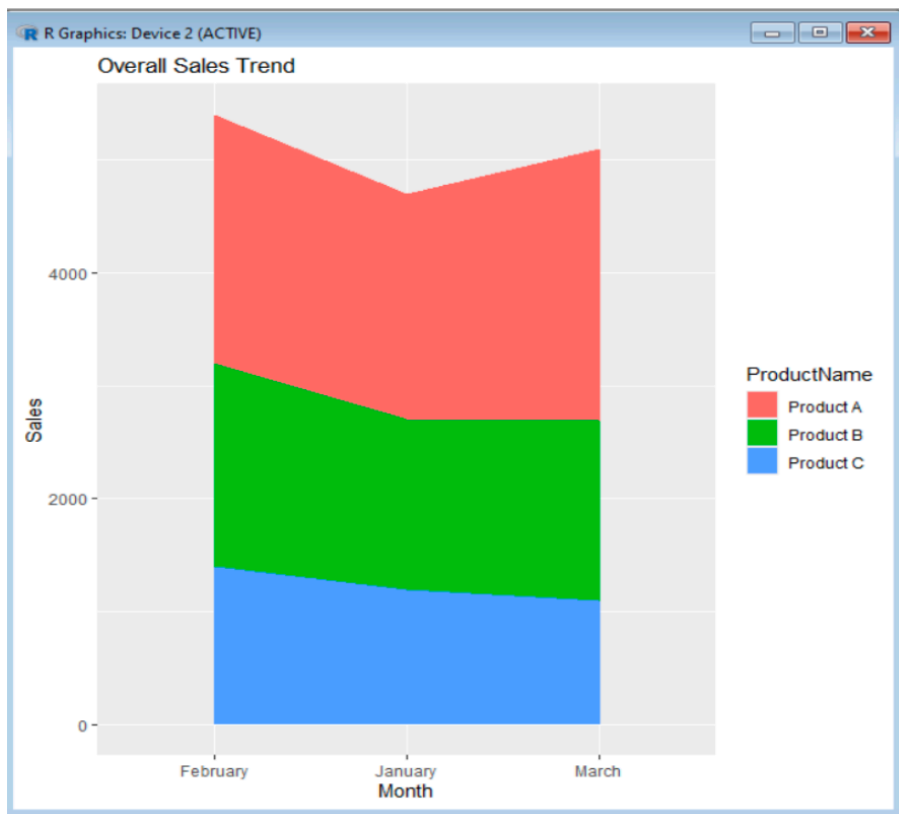


2. Stacked area chart to represent the overall sales trend for all products over the three months:

Python code:

```
sales <- data.frame(  
  ProductID = c(1,2,3),  
  ProductName = c("Product A","Product B","Product C"),  
  January = c(2000,1500,1200),  
  February = c(2200,1800,1400),  
  March = c(2400,1600,1100)  
)  
ggplot(sales_long,  
  aes(x = Month,  
    y = Sales,  
    fill = ProductName,  
    group = ProductName)) +  
  geom_area() +  
  labs(title = "Overall Sales Trend",  
    x = "Month",  
    y = "Sales")
```

Output:



3. Show the monthly sales data for each product:

Code:

```
grid.table(sales)
```

Output:

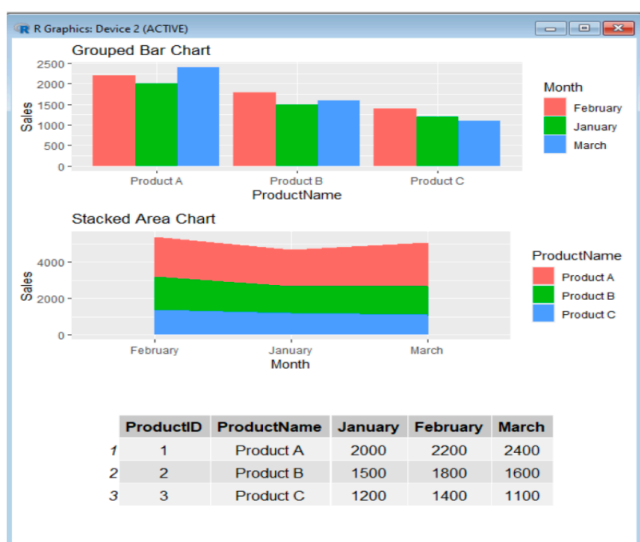
	ProductID	ProductName	January	February	March
1	1	Product A	2000	2200	2400
2	2	Product B	1500	1800	1600
3	3	Product C	1200	1400	1100

4. The grouped bar chart, stacked area chart, and the table for interactive exploration of sales data:

Python code:

```
p1 <- ggplot(sales_long,
aes(ProductName, Sales, fill = Month)) +
geom_bar(stat = "identity",
position = "dodge") +
labs(title = "Grouped Bar Chart")
p2 <- ggplot(sales_long,
aes(Month, Sales,
fill = ProductName,
group = ProductName)) +
geom_area() +
labs(title = "Stacked Area Chart")
grid.arrange(
p1,
p2,
tableGrob(sales),
ncol = 1
)
```

Output:



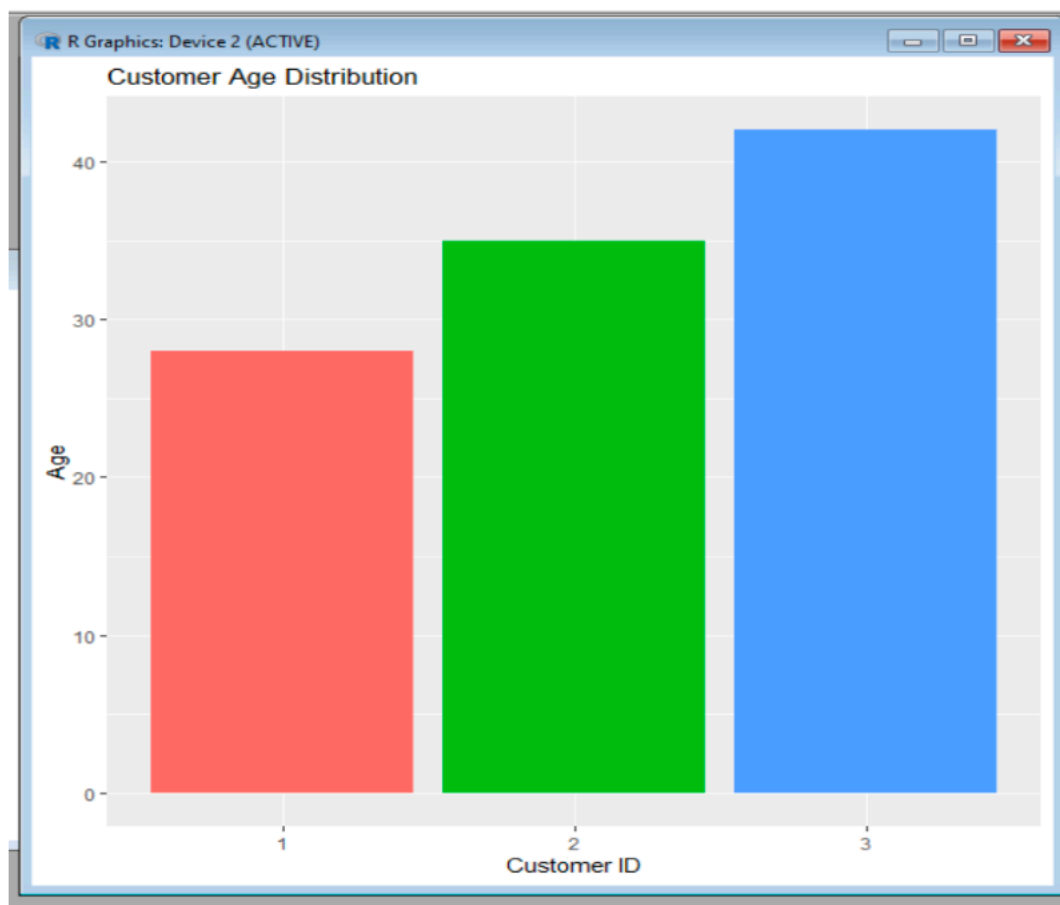
Set 7

1. bar chart to visualize the distribution of customer ages:

python code:

```
customer <- data.frame(  
  CustomerID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)  
ggplot(customer,  
  aes(x = factor(CustomerID),  
    y = Age,  
    fill = factor(CustomerID))) +  
  geom_bar(stat = "identity") +  
  labs(title = "Customer Age Distribution",  
    x = "Customer ID",  
    y = "Age") +  
  theme(legend.position = "none")
```

Output:

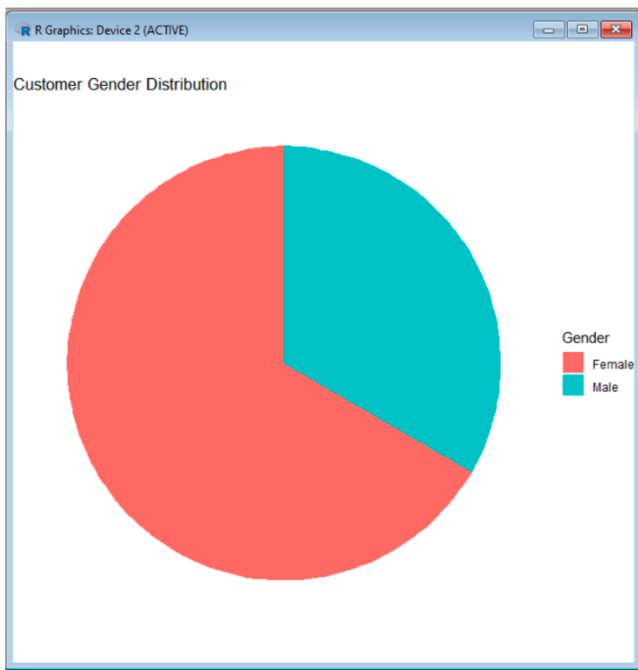


2. Pie chart to represent the distribution of customers by gender:

python code:

```
customer <- data.frame(  
  CustomerID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)  
gender_data <- as.data.frame(table(customer$Gender))  
colnames(gender_data) <- c("Gender", "Count")  
ggplot(gender_data,  
  aes(x = "", y = Count, fill = Gender)) +  
  geom_bar(stat = "identity", width = 1) +  
  coord_polar("y") +  
  labs(title = "Customer Gender Distribution") +  
  theme_void()
```

Output:



3. table to show the demographic information for each customer.

Python code:

```
grid.table(customer)
```

Output:

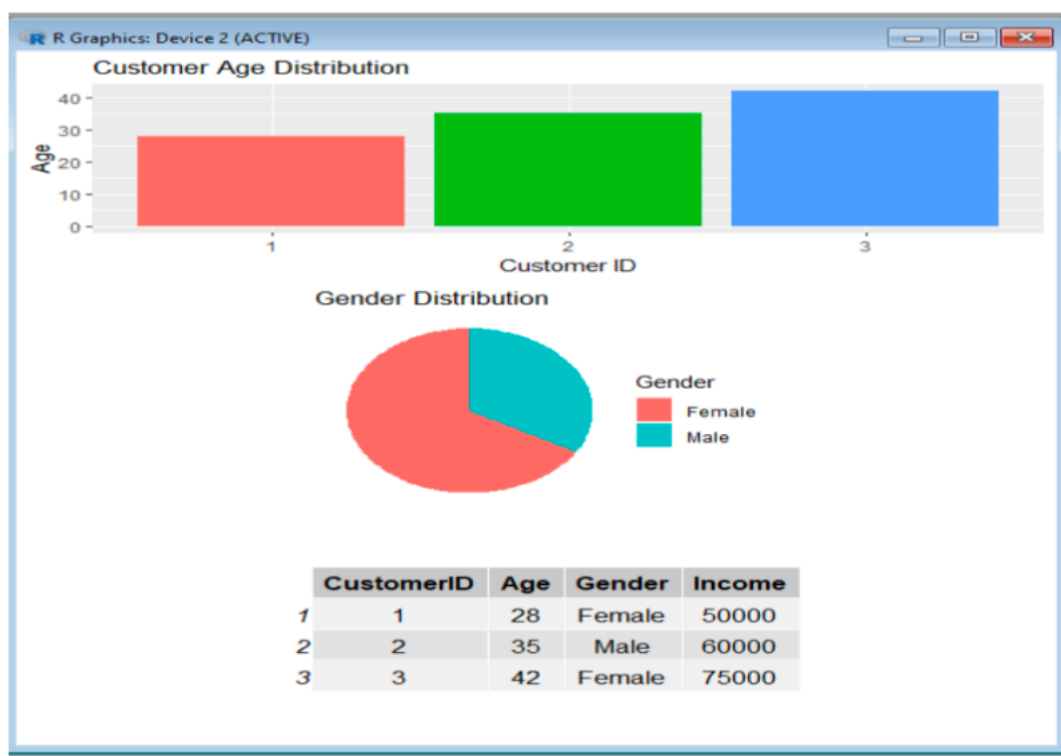
	CustomerID	Age	Gender	Income
1	1	28	Female	50000
2	2	35	Male	60000
3	3	42	Female	75000

4. combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Python code:

```
p1 <- ggplot(customer,
aes(x = factor(CustomerID),
y = Age,
fill = factor(CustomerID))) +
geom_bar(stat = "identity") +
labs(title = "Customer Age Distribution",
x = "Customer ID",
y = "Age") +
theme(legend.position = "none")
gender_data <- as.data.frame(table(customer$Gender))
colnames(gender_data) <- c("Gender", "Count")
p2 <- ggplot(gender_data,
aes(x = "", y = Count, fill = Gender)) +
geom_bar(stat = "identity", width = 1) +
coord_polar("y") +
labs(title = "Gender Distribution") +
theme_void()
grid.arrange(
p1,
p2,
tableGrob(customer),
ncol = 1
)
```

Output:



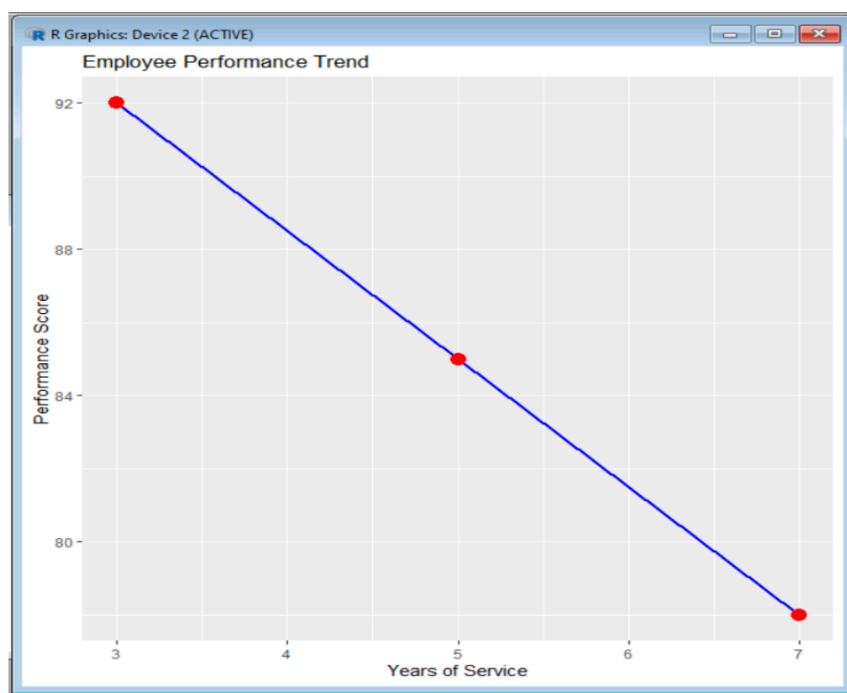
Set 8

1. line chart to visualize the performance trend of employees over time.

Python code:

```
employee <- data.frame(  
  EmployeeID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  YearsService = c(5,3,7),  
  PerformanceScore = c(85,92,78)  
)  
ggplot(employee,  
  aes(x = YearsService,  
    y = PerformanceScore,  
    group = 1)) +  
  geom_line(color = "blue", linewidth = 1) +  
  geom_point(size = 4, color = "red") +  
  labs(  
    title = "Employee Performance Trend",  
    x = "Years of Service",  
    y = "Performance Score"  
  )  
)
```

Output:

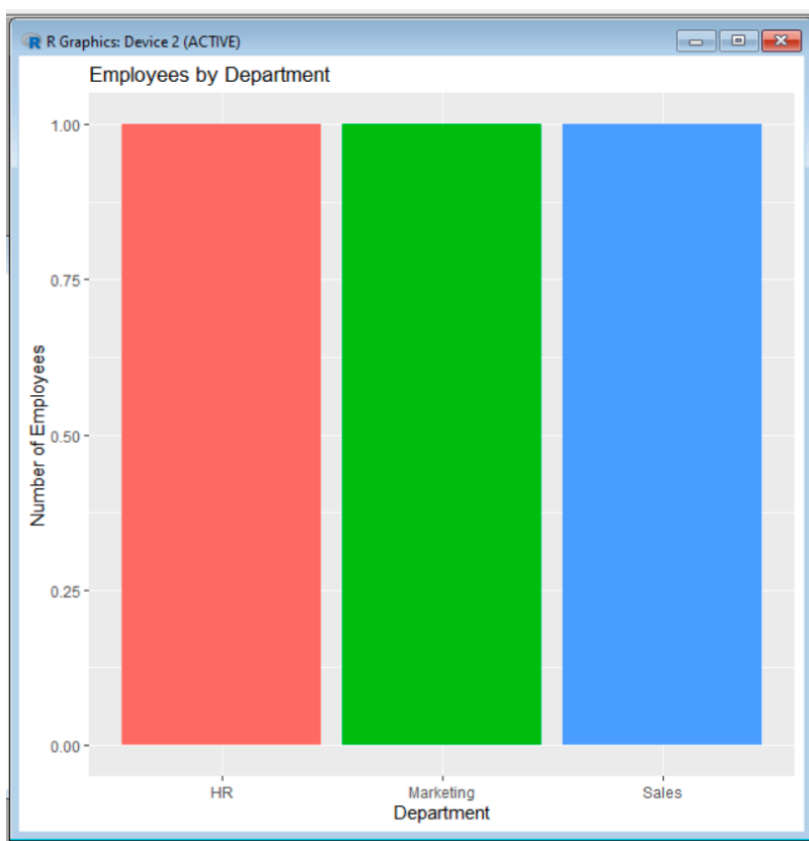


2. bar chart showing the distribution of employees across different departments:

python code:

```
employee <- data.frame(  
  EmployeeID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  YearsService = c(5,3,7),  
  PerformanceScore = c(85,92,78)  
)  
ggplot(employee,  
  aes(x = Department,  
  fill = Department)) +  
  geom_bar() +  
  labs(  
    title = "Employees by Department",  
    x = "Department",  
    y = "Number of Employees"  
  ) +  
  theme(legend.position = "none")
```

Output:

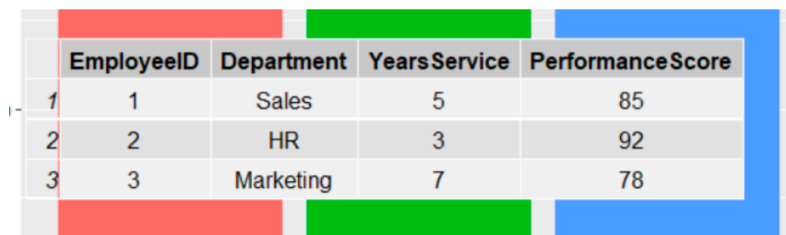


3. table to display the performance data for each employee:

python code:

```
grid.table(employee)
```

output:



	EmployeeID	Department	YearsService	PerformanceScore
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

4. combining the line chart, bar chart, and the table for interactive exploration of employee performance data:

```
library(gridExtra)
```

```
# Line Chart
```

```
p1 <- ggplot(employee,  
aes(x = YearsService,  
y = PerformanceScore,  
group = 1)) +  
geom_line(color = "blue", linewidth = 1) +  
geom_point(size = 4, color = "red") +  
labs(  
title = "Employee Performance Trend",  
x = "Years of Service",  
y = "Performance Score"  
)
```

```
# Bar Chart
```

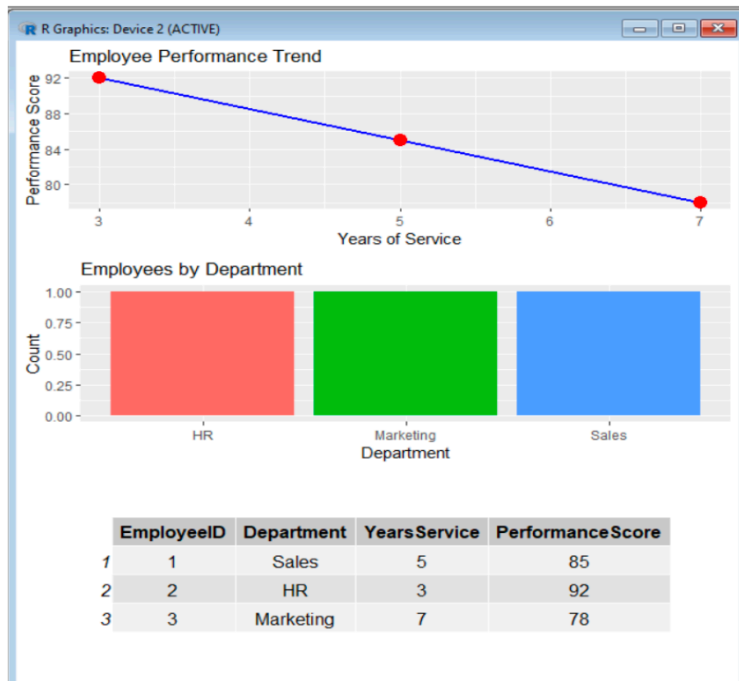
```
p2 <- ggplot(employee,  
aes(x = Department,  
fill = Department)) +  
geom_bar() +  
labs(  
title = "Employees by Department",  
x = "Department",  
y = "Count"  
) +
```

```
theme(legend.position = "none")
```

```
# Dashboard
```

```
grid.arrange(  
p1,  
p2,  
tableGrob(employee),  
ncol = 1  
)
```

Output:



Set 9

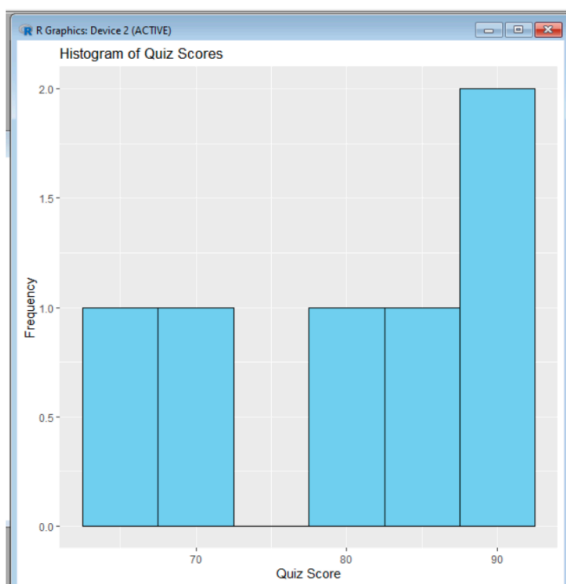
1. histogram for Quiz_Score & boxplot of Quiz_Score by Course.

1.a. Histogram:

Python code:

```
ggplot(data, aes(x = Quiz_Score)) +  
  geom_histogram(binwidth = 5,  
    fill = "skyblue",  
    color = "black") +  
  labs(  
    title = "Histogram of Quiz Scores",  
    x = "Quiz Score",  
    y = "Frequency"  
  )
```

Output:



1.b.Boxplot:

Python code:

```
ggplot(data,  
aes(x = Course,  
y = Quiz_Score,  
fill = Course)) +  
geom_boxplot() +  
labs(  
title = "Quiz Score by Course",  
x = "Course",  
y = "Quiz Score"  
)
```

Output:



2. between Study_Time and Quiz_Score.:

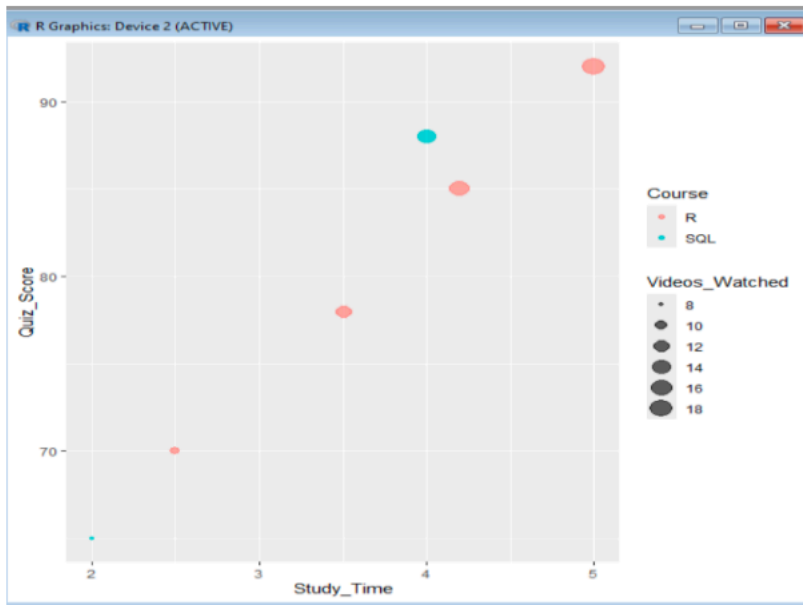
python code:

```
ggplot(data,  
aes(x = Study_Time,  
y = Quiz_Score,  
size = Videos_Watched,  
color = Course)) +  
geom_point(alpha = 0.6) +  
labs(  
title = "Study Time vs Quiz Score",  
x = "Study Time (hrs)",
```

y = "Quiz Score"

)

Output:

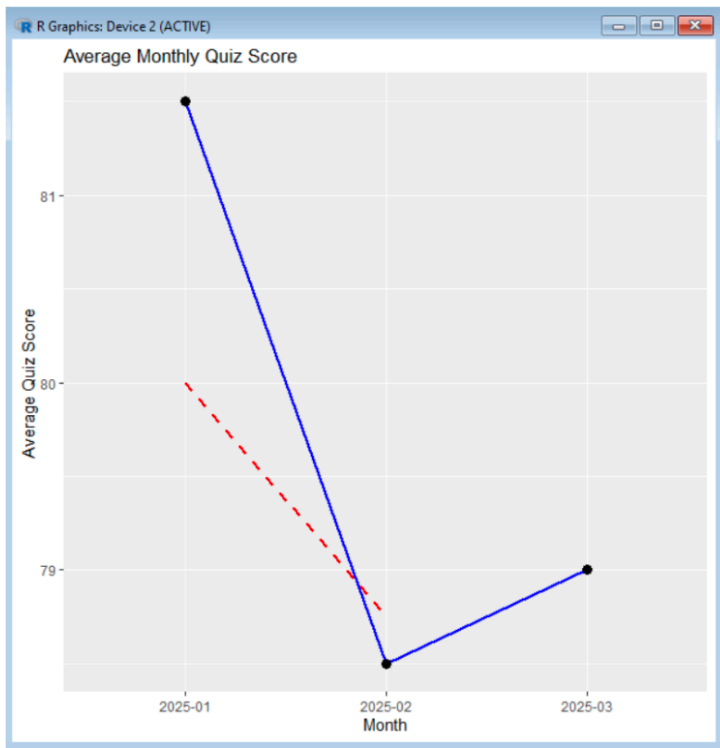


3. convert Login_Date into Date format.

Python code:

```
ggplot(monthly,  
aes(x = Month,  
y = Average_Score,  
group = 1)) +  
geom_line(color = "blue", linewidth = 1) +  
geom_point(size = 3) +  
geom_line(aes(y = MovingAverage),  
color = "red",  
linetype = "dashed",  
linewidth = 1) +  
labs(  
title = "Average Monthly Quiz Score",  
x = "Month",  
y = "Average Quiz Score"  
)
```

Output:



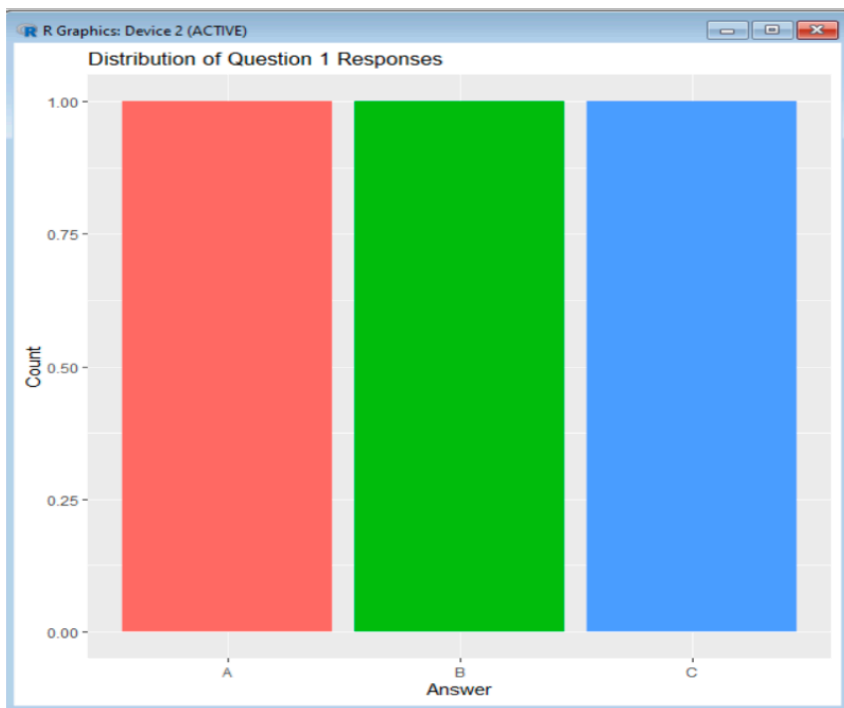
Set 10

1. grouped bar chart to visualize the distribution

python code:

```
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
survey
ggplot(survey,
  aes(x = Question1,
    fill = Question1)) +
  geom_bar() +
  labs(
    title = "Distribution of Question 1 Responses",
    x = "Answer",
    y = "Count"
  ) +
  theme(legend.position = "none")
```

Output:

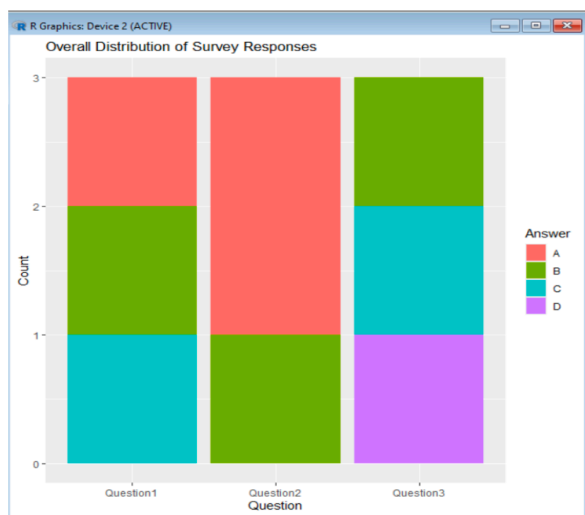


2. stacked bar chart to represent the overall distribution of responses

python code:

```
survey_long <- pivot_longer(
  survey,
  cols = c(Question1, Question2, Question3),
  names_to = "Question",
  values_to = "Answer"
)
ggplot(survey_long,
  aes(x = Question,
  fill = Answer)) +
  geom_bar() +
  labs(
    title = "Overall Distribution of Survey Responses",
    x = "Question",
    y = "Count"
  )
```

Output:

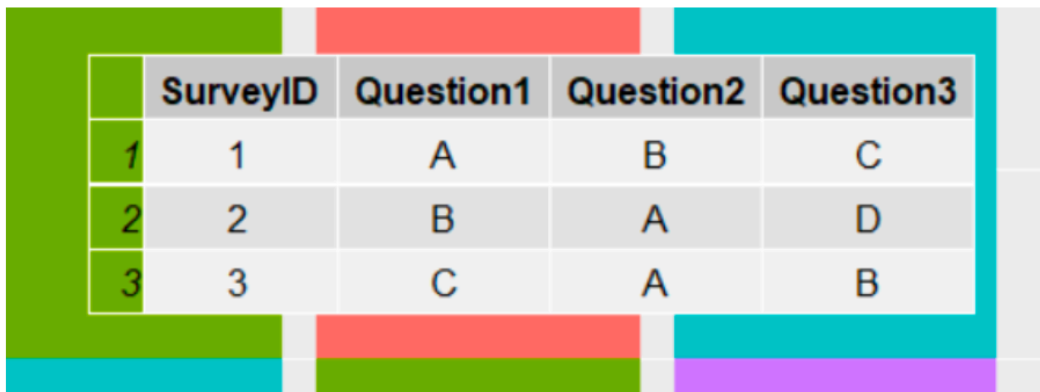


3. table to show the survey response data for each survey.

Python code:

```
grid.table(survey)
```

Output:



	SurveyID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

4. combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Python code:

```
p1 <- ggplot(survey,
aes(x = Question1,
fill = Question1)) +
geom_bar() +
labs(
title = "Question 1 Responses",
x = "Answer",
y = "Count"
) +
theme(legend.position = "none")
survey_long <- pivot_longer(
survey,
cols = c(Question1, Question2, Question3),
names_to = "Question",
values_to = "Answer"
)
p2 <- ggplot(survey_long,
aes(x = Question,
fill = Answer)) +
geom_bar() +
labs(
title = "Overall Survey Responses",
x = "Question",
y = "Count"
)
```

```

grid.arrange(
  p1,
  p2,
  tableGrob(survey),
  ncol = 1
)

```

Output:

